

A Physically Interpretable Single-Mode Phonon-Occupation Model for the Temperature-Dependent Zero-Field Splitting in NV Centers

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Abstract

We present a compact effective model in which the temperature-dependent part of the NV-center zero-field splitting $D(T)$ is governed by the Bose–Einstein occupation of a single representative phonon mode. The model is equivalent to a single-mode coth fluctuation model after absorbing the zero-point contribution into the fitted intercept. Calibrated to the high-precision Ouyang et al. dataset over 298–383 K, the model achieves residuals comparable to empirical polynomial fits while retaining three physically interpretable parameters: the effective intercept D_0 , a coupling amplitude A , and an effective phonon wavenumber $\tilde{\nu}$. The fitted effective wavenumber, $\tilde{\nu} = 711 \pm 48 \text{ cm}^{-1}$, lies within the diamond phonon spectrum and should be interpreted as an effective spin–phonon energy scale rather than a resolved normal mode. When benchmarked against the Liu et al. high-temperature parametrization up to 1000 K, the model remains within approximately 2 MHz over the full range ($R^2 = 0.998185$, RMSE = 1.3 MHz), while a power-law calibration diverges outside the fitting window. We refer to this modelling framework as the *Decoherence-Temperature Quantum Effective Model* (DTQEM), of which the present single-mode thermal shift constitutes the first quantitative component.

1 Introduction

The nitrogen-vacancy (NV) center in diamond is one of the most versatile solid-state quantum sensors [1, 2]. The temperature dependence of the ground-state zero-field splitting, $D(T)$, underpins NV-based nanothermometry [3, 4, 5] and must be accounted for in precision measurements under varying thermal conditions. Although $D(T)$ has been measured over a wide range of temperatures [6, 7, 8, 9, 10], the most common descriptions remain empirical polynomial fits with no direct physical meaning.

A physical interpretation links the temperature shift to lattice expansion and electron–phonon coupling [11]. Recent work by Cambria et al. [12] showed that two effective phonon modes spanning the acoustic and optical branches of the diamond spectrum are sufficient to describe $D(T)$ quantitatively. Building on this idea, we show that a single

effective mode can achieve comparable accuracy within the calibration window and strong extrapolation performance without increasing parameter complexity. This parsimony is attractive because it yields a model with direct physical meaning and a controlled high-temperature asymptote.

We refer to this modelling framework as the **Decoherence-Temperature Quantum Effective Model (DTQEM)**. The present paper constitutes Paper I of this series, addressing the thermal shift of $D(T)$; Paper II will extend the framework to spin-lattice relaxation $1/T_1(T)$ and pure dephasing $\Gamma_\phi(T)$.

2 Model

2.1 Basic form

We model the temperature-dependent zero-field splitting as

$$D(T) = D_0 + A n(\tilde{\nu}, T), \quad (1)$$

where

$$n(\tilde{\nu}, T) = \frac{1}{\exp\left(\frac{hc\tilde{\nu}}{k_B T}\right) - 1} = \frac{1}{\exp\left(\frac{\tilde{\nu}}{k_{B,\text{cm}} T}\right) - 1} \quad (2)$$

is the Bose–Einstein occupation number, $\tilde{\nu}$ is the effective phonon wavenumber (cm^{-1}), and $k_{B,\text{cm}} = k_B/hc = 0.69503476 \text{ cm}^{-1} \text{ K}^{-1}$ is the Boltzmann constant expressed in wavenumber units. Here D_0 is an effective low-temperature intercept, A is a coupling amplitude, and $\tilde{\nu}$ is an effective phonon energy scale.

2.2 Relation to the coth fluctuation form

Although the Bose–Einstein occupation number $n(\tilde{\nu}, T)$ appears naturally as a thermal shift term, it is instructive to note its connection to the more familiar phonon-fluctuation form. The mean-square displacement of a harmonic oscillator mode scales as

$$\langle Q^2 \rangle \propto 2n(\tilde{\nu}, T) + 1 = \coth\left(\frac{hc\tilde{\nu}}{2k_B T}\right), \quad (3)$$

so a general linear coupling gives

$$D(T) = D_{\text{bare}} + \lambda [2n(\tilde{\nu}, T) + 1]. \quad (4)$$

Since the zero-point term λ is temperature-independent, it can be absorbed into the intercept, yielding

$$D(T) = \underbrace{(D_{\text{bare}} + \lambda)}_{D_0} + \underbrace{2\lambda}_A n(\tilde{\nu}, T). \quad (5)$$

The present model (1)–(2) is therefore *equivalent to a single-mode coth fluctuation model*, with the zero-point contribution absorbed into the fitted intercept D_0 . Consequently, D_0 should be interpreted as an effective low-temperature intercept that includes both the bare zero-field splitting and the zero-point lattice contribution, rather than as a precise cryogenic value of $D(0)$.

At high temperature ($k_B T \gg hc\tilde{\nu}$), $n \approx k_B T/hc\tilde{\nu}$ and $D(T)$ becomes approximately linear in T . At low temperature, $D(T)$ approaches D_0 exponentially. The single-mode ansatz is adopted here on grounds of parsimony: it achieves strong fit quality within the calibration window while preserving a physically interpretable structure.

3 Results

3.1 Calibration on high-precision data

We fit the model to the dataset of Ouyang et al. [10], taken from Table 1 of that work (18 points, 298.15–383.15 K, uncertainties $u(D) = (3.6\text{--}8.5) \times 10^{-5}$ GHz). The fitted parameters are

$$D_0 = 2.8764 \text{ GHz}, \quad (6)$$

$$A = -0.1889 \pm 0.0299 \text{ GHz}, \quad (7)$$

$$\tilde{\nu} = 711 \pm 48 \text{ cm}^{-1}. \quad (8)$$

The corresponding fit quality is $R^2 = 0.999946$ with $\text{RMSE} = 1.75 \times 10^{-5}$ GHz.

The Gaussian approximation to the parameter posterior is justified by the small relative uncertainties: $\sigma(D_0)/D_0 < 0.02\%$, $\sigma(A)/|A| \approx 16\%$, and $\sigma(\tilde{\nu})/\tilde{\nu} \approx 7\%$.

The extracted D_0 is slightly lower than the commonly cited $D(0) \approx 2.8775\text{--}2.8780$ GHz [11]. This difference is expected: D_0 is an effective intercept that absorbs the zero-point lattice contribution (see Section 2), and should not be compared directly to cryogenic measurements of $D(0)$ without accounting for that offset.

Parameter estimation and uncertainty propagation

Model parameters were estimated by weighted nonlinear least squares using `scipy.optimize.curve_fit` with the reported standard uncertainties $u(D)$ of Ouyang et al. as 1σ weights (`absolute_sigma=True`), yielding the parameter covariance matrix Σ directly in physical units. The 1σ parameter uncertainties are obtained as $\sigma_i = \sqrt{\Sigma_{ii}}$.

To propagate parameter uncertainty to the model predictions, we drew $N = 10\,000$ independent samples from the joint Gaussian posterior $\mathcal{N}(\hat{p}, \Sigma)$, evaluated $D(T)$ for each sample over the full extrapolation range 280–1020 K, and computed pointwise 2.5th and 97.5th percentiles. The resulting shaded regions in Fig. 2(a) represent *95% parameter-uncertainty bands*; they reflect only the propagated uncertainty in the fitted parameters and do not include measurement noise at the extrapolation temperatures. The same procedure was applied independently to the power-law model to enable a direct visual comparison. The Liu et al. values used for residual evaluation were digitized from Fig. 2(c) of Ref. [14] and carry no independently assigned measurement uncertainty; the ± 5 MHz shading in Fig. 2(b) is a visual guide only.

3.2 Model comparison

All three models achieve comparable fit quality within the calibration window (Table 1). The main distinction is interpretability: the present model uses physically meaningful parameters, whereas the polynomial coefficients are purely empirical.

3.3 Extrapolation benchmark: comparison with Liu et al. up to 1000 K

We apply the parameters obtained from the Ouyang calibration directly to the high-temperature $D(T)$ parametrization of Liu et al. [14]. The values were sampled from the published curve using digital extraction and are used here as a benchmark.

Table 1: Model comparison in the 298–383 K calibration window.

Model	R^2	RMSE (GHz)	Parameters
This work (DTQEM)	0.999946	1.75×10^{-5}	3
Ouyang polynomial	0.999932	1.95×10^{-5}	3
Power law	0.999930	1.98×10^{-5}	3

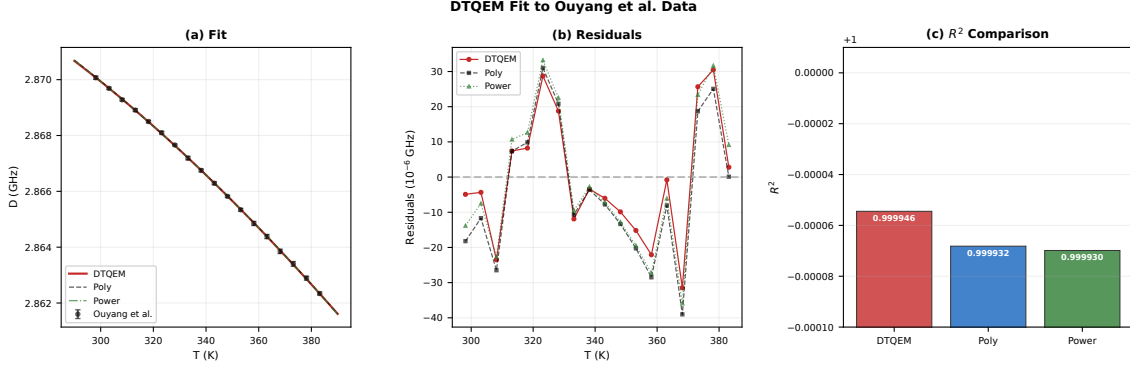


Figure 1: (a) Fit to the Ouyang et al. data. (b) Fit residuals. (c) R^2 comparison. All three models achieve comparable fit quality; RMSE values are below the reported measurement uncertainty.

The deviation remains below 2 MHz across the full range (Table 2). In contrast, a power-law model exhibits deviations that grow rapidly outside the fitting window. Over the full 300–1000 K range, the DTQEM extrapolation yields $R^2 = 0.998185$ and an RMSE of 1.32 MHz. The Liu et al. parametrization lies within the DTQEM 95% parameter-uncertainty band across the full 300–1000 K range, whereas the power-law band diverges above approximately 600 K.

Table 2: Comparison with the Liu et al. parametrization up to 1000 K.

T (K)	D_{Liu} (GHz)	D_{model} (GHz)	Difference (MHz)
300	2.8680	2.8699	−1.9
400	2.8600	2.8605	−0.5
500	2.8490	2.8484	+0.6
600	2.8360	2.8344	+1.6
700	2.8210	2.8194	+1.6
800	2.8050	2.8035	+1.5
900	2.7880	2.7872	+0.8
1000	2.7700	2.7704	−0.4

4 Discussion

4.1 Physical interpretation

The fitted wavenumber $\tilde{\nu} = 711 \pm 48 \text{ cm}^{-1}$ ($\approx 88 \text{ meV}$) lies between the acoustic-phonon regime and the zone-center optical phonon at 1332 cm^{-1} , and is bracketed by the two

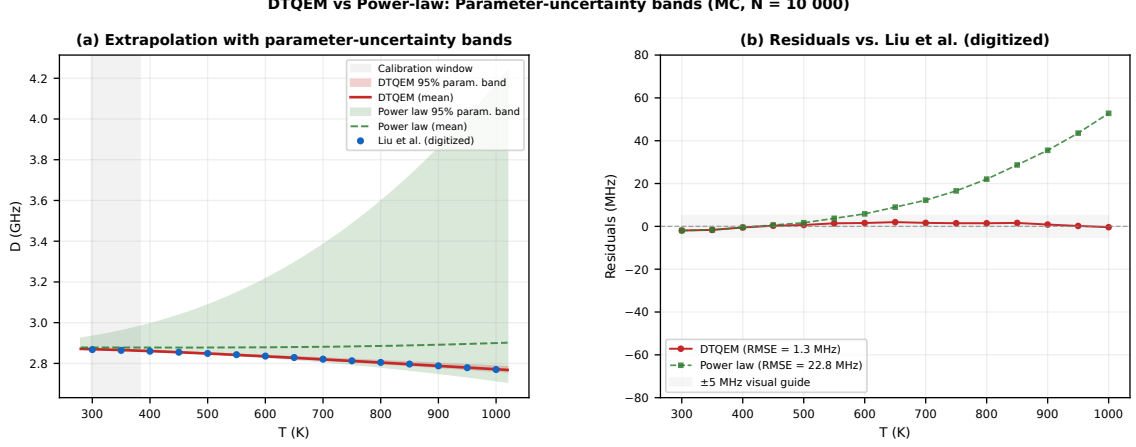


Figure 2: Extrapolation of the DTQEM and power-law models from the 298–383 K calibration window (gray shading) to the Liu et al. parametrization up to 1000 K. (a) Model predictions with 95% parameter-uncertainty bands obtained by Monte Carlo sampling ($N = 10\,000$). (b) Residuals with respect to the digitized Liu et al. values; the ± 5 MHz gray band is a visual guide.

effective modes reported by Cambria et al. [12]. It should therefore be interpreted as an effective energy scale representing the dominant spin–phonon coupling channel, not as a spectroscopically resolved single lattice mode.

A Debye-integral model was also tested as an alternative, replacing $n(\tilde{\nu}, T)$ with the integrated phonon density of states weighted by a Debye cutoff frequency $\tilde{\nu}_D$. Although physically motivated, this model requires numerical integration, introduces an additional free parameter, and yields no measurable improvement in R^2 or RMSE within the calibration window. The single-mode ansatz is therefore retained on grounds of parsimony.

4.2 Identifiability and scope

The model should be interpreted as a compact effective description whose individual parameters are constrained by the present dataset but not fully uniquely identifiable without broader temperature coverage. The agreement with the Liu et al. parametrization indicates a robust thermal trend rather than a fitting artifact. The model should be used with caution above 1000 K, where higher-order phonon–phonon interactions and anharmonic effects become significant.

The uncertainty on the effective phonon wavenumber, $\sigma(\tilde{\nu}) = 48 \text{ cm}^{-1}$ ($\sim 7\%$), could be substantially reduced by incorporating low-temperature measurements (e.g., $T < 100 \text{ K}$), where the exponential sensitivity of $n(\tilde{\nu}, T)$ to $\tilde{\nu}$ is strongest. Alternatively, a joint fit to both $D(T)$ and $1/T_1(T)$ over a common temperature range would provide independent constraints on the coupling amplitude A and the effective wavenumber $\tilde{\nu}$, improving parameter identifiability without requiring new instrumentation beyond what is standard in NV-center spectroscopy.

4.3 Implications for thermometry

The model combines a physical low-temperature limit, a controlled high-temperature asymptote, and strong extrapolation performance, making it preferable to unconstrained

polynomial fits for NV thermometry applications.

4.4 Possible extension to relaxation and dephasing (Paper II)

The same phonon-occupation framework may be extended to spin-lattice relaxation rates $1/T_1(T)$ and pure dephasing $\Gamma_\phi(T)$. For pure dephasing, the natural ansatz is:

$$\Gamma_\phi(T) = \Gamma_0 + \alpha [2n(\tilde{\nu}_\phi, T) + 1] = \Gamma_0 + \alpha \coth\left(\frac{hc\tilde{\nu}_\phi}{2k_B T}\right), \quad (9)$$

while for spin-lattice relaxation:

$$\frac{1}{T_1(T)} \propto n(\tilde{\nu}, T) [n(\tilde{\nu}, T) + 1]. \quad (10)$$

A key question for Paper II is whether the same effective wavenumber $\tilde{\nu}$ governs both $D(T)$ and $\Gamma_\phi(T)$ (single dominant spin-phonon channel), or whether distinct modes must be introduced. A robust quantitative test would require experimental data spanning at least 100–600 K with sub-percent precision on T_1 . This is left as a concrete objective for Paper II of this series.

5 Conclusion

We have presented a physically motivated single-mode occupation-number model for $D(T)$ in NV centers, and shown its equivalence to a single-mode coth fluctuation model with an absorbed zero-point term. The model matches the best published polynomial fit within the reported experimental uncertainties of the Ouyang et al. dataset while providing direct physical interpretability. The extracted effective wavenumber $\tilde{\nu} = 711 \pm 48 \text{ cm}^{-1}$ is consistent with the diamond phonon spectrum. Most importantly, when calibrated only on 298–383 K, the model reproduces the Liu et al. parametrization up to 1000 K with $R^2 = 0.998185$, whereas a power-law calibration performs poorly outside the fitting window. These results establish DTQEM as a compact, physically interpretable alternative to purely empirical descriptions of $D(T)$ and lay the groundwork for a unified spin-phonon framework connecting thermal shifts and decoherence rates in NV centers.

Data Availability

The Ouyang et al. data used for model calibration are available in Table 1 of Ref. [10]. The Liu et al. parametrization values used for the extrapolation benchmark were digitally extracted from Fig. 2(c) of Ref. [14] and are tabulated in the present manuscript. A preprint version of this work, along with the Python fitting code and the digitized benchmark values, is available at DOI: [<https://doi.org/10.5281/zenodo.20819966>].

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Conflict of Interest

The author declares no conflict of interest.

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